

Lecture 18 Scribe – Marginal PMF, Correlation, Covariance, Joint Gaussian RVs

(Exam Preparation Notes – Step-by-Step Reasoning)

1. Marginal PMF

1.1 Definition

The **marginal PMF** of one variable is obtained by **summing the joint PMF** over all possible values of the other variable.

Step-by-step reasoning

1. Start with a **joint PMF** ($P(X, Y)$), which gives probabilities of pairs $((x, y))$.
2. Suppose we are interested only in **one variable**, say (X) .
3. The probabilities in the joint PMF include the influence of (Y) .
4. To isolate (X) , we:
 - **Fix** $(X = x)$
 - **Sum over all possible values of** (Y)

Thus, we remove the dependency on (Y) by aggregation.

1.2 Marginal PMF from Joint PMF Table

Procedure

Given a table of $(P(X, Y))$:

To compute $(P_X(x))$:

1. Select a fixed value (x)
2. Identify all entries where $(X = x)$
3. Sum all corresponding probabilities across different (y)

To compute $(P_Y(y))$:

1. Fix (y)
2. Sum all entries across different (x)

1.3 Example (Logical Derivation)

Given a joint PMF:

Solution steps

1. List all values of (X) and (Y)
2. For each value of (X) :
 - Add probabilities across all (Y)
3. For each value of (Y) :
 - Add probabilities across all (X)

Conclusion

The resulting sums form:

- Marginal PMF of (X)
- Marginal PMF of (Y)

2. Correlation

2.1 Definition (Product Moment)

A basic way to measure joint behavior is through the product moment.

It computes:

- The average value of the product (XY)

2.2 Interpretation

Step-by-step understanding

1. For each outcome:
 - Multiply (X) and (Y)
2. Take expectation:
 - This gives $(E[XY])$

3. This reflects:

- Whether large values of (X) occur with large values of (Y)

2.3 Limitations

The measure:

$$R_{XY} = E[XY]$$

Issues (step-by-step)

1. **Depends on scale**

- If (X) or (Y) is scaled, value changes

2. **Computed about origin**

- Does not consider mean values

3. **Mixes multiple effects**

- Mean
- Spread
- Dependence

Special case

$$R_{XY} = 0 \Rightarrow \text{orthogonal about the origin}$$

2.4 Transition to Covariance

To fix these issues:

1. Remove the effect of mean

2. Center variables:

- Replace (X) with $(X - E[X])$
- Replace (Y) with $(Y - E[Y])$

This leads to **covariance**.

3. Covariance

3.1 Definition

We first subtract the mean from each variable and then measure how the centered quantities vary together.

3.2 Step-by-step Derivation

1. Compute mean values:

- $(E[X]), (E[Y])$

2. Form deviations:

- $(X - E[X])$
- $(Y - E[Y])$

3. Multiply deviations:

- $((X - E[X])(Y - E[Y]))$

4. Take expectation:

- This gives covariance

3.3 Interpretation

Covariance captures:

- **Positive covariance:**
 - Both variables deviate in same direction
- **Negative covariance:**
 - Variables deviate in opposite directions

3.4 Algebraic Form (Step-by-step)

Start with definition:

1. Expand product:

$$(X - E[X])(Y - E[Y])$$

2. Multiply terms

3. Apply expectation

4. Use linearity of expectation

5. Simplify

This gives a computationally simpler expression.

3.5 Properties of Covariance

1. Symmetry

$$\text{Cov}(X, Y) = \text{Cov}(Y, X)$$

2. Variance as Special Case

$$\text{Var}(X) = \text{Cov}(X, X)$$

3. Linearity w.r.t Constants

Scaling variables scales covariance accordingly

4. Independence

If (X) and (Y) are independent:

$$\text{Cov}(X, Y) = 0$$

Important conclusion

- Zero covariance $\Rightarrow$ **uncorrelated**
- But **does NOT** imply independence

3.6 Example 1 (Step-by-step)

Given

Expressions for (X) , (Y)

Solution steps

1. Write covariance formula
2. Substitute given expressions
3. Expand terms
4. Apply expectation operator
5. Simplify step-by-step

Conclusion

Final value obtained after systematic substitution

3.7 Example 2 (Continuous Case)

Given a joint PDF.

Step-by-step solution

1. Compute:
 - $(E[X])$ using integration
2. Compute:
 - $(E[Y])$
3. Compute:
 - $(E[XY])$
4. Substitute into covariance formula
5. Simplify result

4. Pearson's Correlation Coefficient

4.1 Definition

Pearson's correlation coefficient calculates the effect of change in one variable when the other variable changes.

4.2 Step-by-step Interpretation

1. Start from covariance
2. Normalize using:
 - Standard deviation of (X)
 - Standard deviation of (Y)
3. This removes unit dependence

4.3 Properties

Measures:

- **Strength** of relationship
- **Direction** of relationship

It is:

- **Unit-free**
- **Normalized covariance**

5. Jointly Gaussian Random Variables

5.1 Definition

A pair $((X, Y))$ is jointly Gaussian if:

- The joint density has bivariate Gaussian form
- Marginals are Gaussian

5.2 Intuition

Step-by-step reasoning

1. Many real-world variables are influenced by:
 - Multiple small random factors
2. When combined:
 - They tend to follow Gaussian behavior
3. Hence, joint distribution becomes Gaussian

5.3 Applications

- Signal processing
- Communication systems
- Noisy measurements

5.4 Example: Climate System

Variables:

- (X) : CO₂ concentration
- (Y) : Temperature anomaly

Reasoning

1. Both depend on multiple random processes:
 - Weather
 - Ocean circulation
 - Emissions
2. These combined uncertainties:

- Produce approximately Gaussian behavior
3. Hence:
- Joint distribution $\approx$ Gaussian

5.5 Algebraic Form and Interpretation

Joint Gaussian is characterized by:

1. Means
2. Variances
3. Covariance

These parameters fully describe:

- Shape
- Spread
- Dependence structure

Final Exam Summary

Core Flow of Concepts

1. Joint PMF $\rightarrow$ Marginal PMF (by summation)
2. Raw correlation ($E[XY]$) $\rightarrow$ not centered
3. Center variables $\rightarrow$ Covariance
4. Normalize covariance $\rightarrow$ Pearson coefficient
5. Extend to continuous case $\rightarrow$ Joint Gaussian RVs

Key Insights to Remember

- Marginalization removes dependence on one variable
- Covariance measures **joint deviation from mean**
- Zero covariance $\neq$ independence
- Pearson coefficient is **scale-free**
- Joint Gaussian models real-world continuous systems